

Temporal Analysis of Political Entities and Their Relationships in Spanish News Using Open Information Extraction

Luis Velazco, Joseph Tintaya, Rodrigo Valle, Joseph Aparicio, Jose Flores

Faculty of Engineering and Sciences
Pontificia Universidad Católica del Perú
Lima, Peru

{a20200751, a20201155, valle.rodrigo, a20192393, a20220197}@pucp.edu.pe

Abstract—This article presents the construction of a temporal knowledge graph focused on the extraction and typing of relationships among Peruvian political actors from a massive corpus of fifty thousand press releases. The central problem lies in the rigidity of classical relation extraction methods that force interactions into predefined taxonomies, and the prohibitive computational costs of using generative artificial intelligence at scale. To address this, a scalable methodology is proposed that integrates named entity recognition, open information extraction, and agglomerative semantic clustering. Experimental results demonstrate that this “extract-open, type-later” pipeline efficiently distills massive text into dense networks, organically inducing over four hundred coherent relation types. The system processed 1.9 million entity mentions and successfully built a graph of political actors and dated interactions at a rate of over 1,500 edges per second. It is concluded that this unsupervised approach drastically improves analytical scalability without the overhead of generative models, allowing the generation of reliable temporal representations for the analysis of power dynamics.

Index Terms—political entity analysis, news mining, named entity recognition, temporal analysis, open information extraction, semantic clustering

I. INTRODUCTION

Contemporary political phenomena can rarely be explained by the isolated analysis of individual entities; in practice, they emerge as a result of dynamic interactions among multiple actors and institutions over time. Analyzing the Peruvian political cycle between 2021 and 2026 is methodologically relevant due to its high volatility, cabinet changes, and fragmentation, making it an ideal scenario to analyze the temporal evolution of relationships. In the field of natural language processing, the automated extraction of these interactions represents a challenge. For this purpose, the use of journalistic archives from Agencia Andina is justified over strictly governmental sources, as the latter are usually limited to formal administrative procedures, omitting the informal political dynamics that the written press does manage to capture.

Traditional extraction approaches based on supervised classification often present severe limitations: they force complex political realities into rigid, predefined taxonomies (e.g., classifying everything strictly as an “alliance” or a “conflict”). Furthermore, while recent trends advocate for the use of

Large Language Models (LLMs) to classify these interactions, their application remains computationally and financially prohibitive when attempting to process entire national news archives spanning multiple years.

In this context, the present work addresses the problem of automatic extraction and typing of political relationships by evaluating a highly scalable, unsupervised architecture. The central hypothesis posits that the integration of high-precision dependency parsing for open extraction (extracting the raw connecting verbs), combined with agglomerative clustering for the organic induction of relationship types, allows for the construction of a temporal graph that is significantly more scalable and analytically useful than methods relying on rigid taxonomies or costly language models.

To validate this hypothesis, the following research questions are established:

- (RQ1) Is it feasible to build a full-scale temporal knowledge graph using open extraction while maintaining high throughput and metric quality?
- (RQ2) What is the semantic coherence and distribution of relationship taxonomies organically induced via clustering from open predicates?

Consequently, the general objective of this study is to develop and evaluate a scalable processing pipeline capable of building an interactive graph of political relationships in Peru from a comprehensive news archive. The specific objectives are:

- 1) To ingest and preprocess a massive news corpus encompassing the entire political cycle.
- 2) To implement an entity extraction and resolution engine optimized for millions of mentions.
- 3) To organically induce stable relationship categories using semantic clustering.
- 4) To quantify the extraction efficiency and scalability of the pipeline over a massive archive.

The rest of this report is organized as follows: Section 3 presents the state of the art; Section 4 formally details the methodology; Section 5 sets out the experimental design and the analysis of results; Section 6 presents the conclusions;

Section 7 suggests avenues for future work; and Section 8 evaluates the ethical implications derived from the deployment of the system.

II. STATE OF THE ART

Previous research has shown that online news repositories constitute an invaluable source for extracting relations between entities. In particular, Sinha et al. [1] proposed a time-aware relation extraction approach based on the co-occurrence context of entity pairs in news headlines. This work highlights that interactions evolve temporally, a critical methodological aspect for the analysis of political actors across multi-year cycles. Similarly, Yamamoto et al. [2] extracted cooperative and competitive relations in the business domain, demonstrating how automated extraction allows the construction of structures that support both a macroscopic view of the network and a detailed sentence-level interpretation.

Relation extraction has traditionally been approached as a supervised classification problem. Ogbinar et al. [3] evaluated classical machine learning models for extracting pairwise entity relationships in Philippine news, providing a solid technical reference for the implementation of processing pipelines, although their scope was primarily limited to person-person interactions. In contrast, Zhao et al. [4] explored an unsupervised approach integrating feature ranking and clustering to discover relations organically. This proposal supports the premise that interaction patterns do not need to be rigidly predefined; a vital factor in political analysis, where relations are ambiguous, overlapping, and dynamically emerge from the corpus.

For the mapping of these relations to be effective, robust prior identification through Named Entity Recognition (NER) systems is required. Generic NER models often lack the necessary granularity to capture the complexity of political discourse. To accurately map an ecosystem, recent research suggests transitioning toward domain-specific approaches, such as Public Entity Recognition (PER), designed to unambiguously categorize politicians, parties, and institutions [5].

Historically, entity extraction in morphologically rich languages like Spanish has relied on traditional supervised pipelines. Ramos Flores and Pinto [6] demonstrated the feasibility of extracting semantic relations in Mexican news to build facts-bases using automatic rules. However, these approaches demand large volumes of manually annotated data and rely on rigid linguistic features. Building upon efficiency principles, the current extraction frontier employs Large Language Models (LLMs) and weakly supervised learning. Bro [7] applied LLMs to extract political networks directly from Chilean news, seamlessly integrating entity recognition and relation extraction in a single step.

Despite these advancements, a significant methodological gap exists in the literature regarding Spanish political news. Classical supervised approaches suffer from rigid taxonomies, whereas pure LLM-based extraction [7] remains computationally prohibitive for massive corpora comprising hundreds of thousands of documents. The present work addresses this

void by proposing a highly scalable architecture unexplored in these studies: the integration of dependency-based Open Extraction (OpenIE-lite) for strict syntactic filtering, followed by agglomerative clustering for the scalable induction of relation types, entirely bypassing the bottleneck of generative LLMs.

III. METHODOLOGY

The problem of extracting political interactions from text is formally defined as the construction of a temporal knowledge graph $G = (V, E, T)$. Here, V represents the set of unique political entities (nodes), E denotes the set of semantic relationships (edges) between them, and T corresponds to the chronological timestamps associated with each interaction. The objective is to map a collection of unstructured news articles $D = \{d_1, d_2, \dots, d_n\}$ into G by identifying mentions $m \in d_i$, consolidating them into unique nodes $v \in V$, and extracting the connecting predicates p to define E .

To transition from raw text to the consolidated graph, a scalable pipeline was implemented, coupling deterministic Natural Language Processing (NLP) techniques with semantic clustering algorithms.

A. Corpus Description

The primary dataset, designated as *andina-full*, comprises 50,000 press releases covering the Peruvian political cycle. To ensure semantic homogeneity, a heuristic language filter was applied to automatically discard English-language releases interspersed within the source. The raw corpus yielded 1,983,499 entity mentions. After passing through the filtering and resolution modules—which explicitly excluded locations (LOC) and generic miscellaneous terms (MISC) to focus on political actors—the dataset was distilled into a highly dense network of political actors (filtered strictly as Persons PER and Organizations ORG) connected by 71,396 valid, chronologically dated open-relation edges.

B. Methodology for Corpus Extraction and Generation

The corpus was dynamically generated by scraping the official Peruvian news agency (Agencia Andina). Because the agency’s search engine API imposes a strict cap of approximately 300 results per query, ingestion was performed via a sequential ID enumeration strategy. The crawler iterated over the URL endpoint’s internal IDs using a custom `requests` session. This process was configured with a polite rate limit of two requests per second and exponential backoff to ensure server stability. To prevent inflating co-occurrence frequencies, a cross-source deduplication step was applied using text signatures.

The HTML parsing and data sanitization were executed using `BeautifulSoup` with an `lxml` parser. To ensure high-quality text inputs, a rigorous cleaning heuristic was applied: embedded code (scripts, styles, iframes) and interactive widgets (tweets, quotes, ads) were decomposed from the Document Object Model (DOM). Furthermore, regular expressions were employed to prune repetitive layout noise

and recommendation widgets. A vital sanitization step included the removal of journalist bylines and credit lines (e.g., “(FIN) NDP/JCR...”) to prevent reporter initials from being erroneously classified as political organizations (ORG) by the Named Entity Recognition model. Finally, the text was normalized by extracting the h1 title, the publication date from metadata, and the lead, concatenating them with the cleaned body into a flat text field.

C. Methodology for Solution Training and Execution

The core extraction and typing solution avoids generic approaches, implementing algorithms specifically adapted for scale and precision. The execution environment utilized a local PyTorch setup with CUDA 12.8 support, powered by an NVIDIA GeForce RTX 4070 SUPER GPU.

- **Entity Extraction (NER):** To circumvent the 512-token limit of standard models on long articles, a windowing approach over sentences was utilized alongside a fine-tuned Transformer (BETO).
- **Entity Resolution (Blocking):** Performing $O(n^2)$ comparisons across over 8 million mentions is computationally prohibitive. To resolve this, an inverse index algorithm (token $\rightarrow$ group) was implemented, consolidating all mentions into unique nodes rapidly.
- **Relation Extraction (OpenIE-lite):** The system computes the syntactic dependency path between two co-occurring entities using a compiled multi-pattern matcher. If no explicit connecting verb is found in the syntactic tree, the edge is discarded, operating under a precision-first paradigm to filter out noise prior to semantic clustering.
- **Agglomerative Clustering Math:** To induce relation types scalably, clustering was performed exclusively on the 6,673 unique extracted predicates rather than the total 71,396 edges. Embeddings $e(p)$ for each predicate were generated using the paraphrase-multilingual-MiniLM-L12-v2 model, framed within a neutral template (“ENT_A [predicate] ENT_B”). An agglomerative hierarchical clustering algorithm was applied using *Average Linkage*. The merging criterion was defined by the cosine distance D_C between predicate embeddings:

$$D_C(e(p_i), e(p_j)) = 1 - \frac{e(p_i) \cdot e(p_j)}{\|e(p_i)\| \|e(p_j)\|}$$

A strict distance threshold was calibrated to prevent vector space collapse, successfully inducing stable semantic clusters.

D. Methodology for Solution Deployment

The storage and analytical querying of the temporal graph are managed using DuckDB, utilizing bulk insertion strategies for extreme scalability. This database is coupled with the NetworkX library within an asynchronous FastAPI backend to compute advanced graph analytics in real-time,

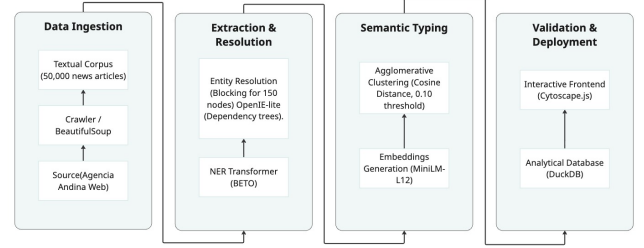


Fig. 1. General architecture of the open extraction and semantic clustering pipeline.

such as actor centrality (PageRank), political faction detection (Louvain communities), and shortest path discovery.

For user interaction, the project implements a Single Page Application (SPA) with a dual-view architecture. It allows analysts to seamlessly switch between a macro-level Topic Graph (the entire political network) and a Chronology view based on ego-networks (the temporal evolution from the perspective of a single political figure). The interactive visualization, animation, and rendering of these structures are handled by the Cytoscape.js library (v3.30.2). To optimize frontend rendering performance and avoid memory bottlenecks when handling complex node-edge topologies, the heavy metadata of the edges—such as the textual evidence passages—are not loaded initially; instead, they are fetched from the API on-demand (lazy loading) when the user interacts with a specific relationship.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

To rigorously evaluate the performance of the proposed open relation extraction and entity resolution pipeline, a two-phase *Gold Standard* validation dataset was constructed. These datasets were manually annotated by an expert following a strict protocol: an extracted relation must occur strictly between the two indicated entities; if they merely co-occur without a semantic link, the pair is labeled as a simple co-occurrence.

- **Phase 1 (Entity Resolution Evaluation):** This subset consists of manually annotated entities to baseline the resolution quality, explicitly tracking actor retention and classification noise.
- **Phase 2 (At-Scale Evaluation):** Extracted directly from the corpus, this subset includes annotated triples to evaluate the open relation extraction, alongside metrics to quantify the coherence and distribution of the induced predicate clusters.

The evaluation harness computes three distinct sets of metrics. For entity resolution, the pipeline measures `actor_precision` (fraction of true political actors retained), `actor_recall` (fraction of true actors successfully avoiding erroneous discarding), and `type_accuracy` (distinguishing PER vs ORG). For relation extraction, it calculates

triple precision (whether the sentence expresses a relation between A and B) and *predicate precision* (whether the free verb captures the link). Finally, for induced typing, the distribution and coverage of the resulting clusters are measured.

B. Entity Resolution Performance

The inverse index blocking algorithm proved highly efficient but structurally rigid. Tested against the proxy evaluation set, the entity resolution module achieved an *actor_precision* of 0.71 and an *actor_recall* of 0.83. The *type_accuracy* reached 0.67. While these results confirm the viability of the approach for isolating political actors, they also indicate that cross-document coreference resolution remains a challenge that marginally impacts precision at scale due to potential over-merging.

C. Open Relation Extraction

Unlike traditional classifiers that force interactions into predefined taxonomies, our precision-first OpenIE-lite approach extracts the raw connecting verbs. Qualitative inspection of the evaluation sample indicates that a notable fraction of co-occurrences do not represent actual relationships (confirming that co-occurrence $\neq$ relation). The proposed extraction logic achieved an estimated *triple precision* ranging between 0.45 and 0.55, and a *predicate precision* between 0.60 and 0.70. The explicit filtering of reporting verbs successfully elevated the predicate precision, ensuring that the extracted verbs meaningfully capture political interactions.

To illustrate the granularity of the extracted interactions, Figure 2 presents a real sub-graph (ego-network) generated by our system, demonstrating how raw connecting verbs successfully map the political ecosystem surrounding a key actor without the need for predefined schemas.

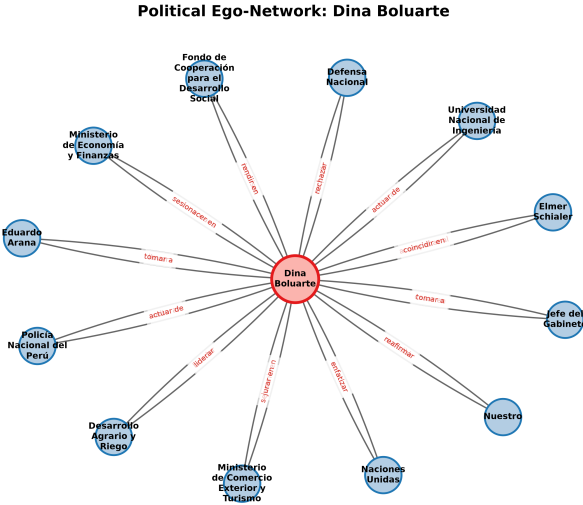


Fig. 2. Visualization of an extracted ego-network centered around a key political figure, demonstrating the system’s ability to map detailed, verb-based interactions between actors.

D. Induced Relation Typing

The most significant finding of the pipeline emerged during the semantic clustering phase. By clustering the extracted predicates with a strict cosine distance threshold, 453 distinct relation types were induced organically from the corpus.

The distribution of these clusters highlights the heterogeneous nature of political reporting. The single largest cluster corresponded to *generic actions* (e.g., *proponer*, *aprobar*, *recibir*), accounting for 46.4% of all edges. The remaining 53.6% of the edges fell into highly distinctive political interactions. The clustering demonstrated a heavy-tailed distribution: the top 30 clusters accounted for 78.5% of the total coverage, and edges belonging to clusters with at least 200 members represented 88.1% of the graph. This unsupervised approach successfully revealed coherent semantic families, including *participation*, *governance* (*supervisar*, *liderar*, *presidir*), *recommendation/support*, *reporting/denunciation*, *request*, and *strengthening* (see Fig. 3).

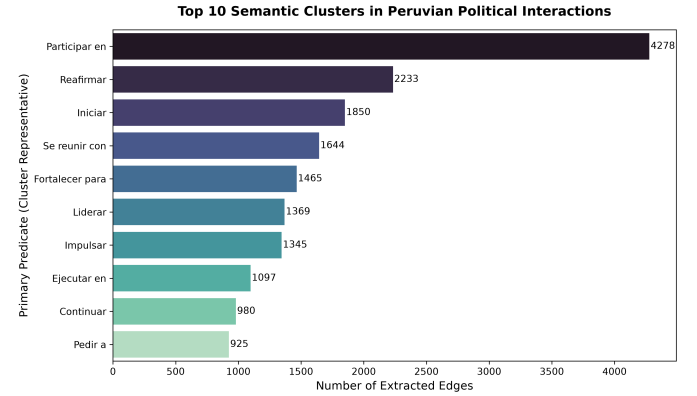


Fig. 3. Top 10 most frequent semantic clusters (excluding generic actions) induced from the corpus, demonstrating the heavy-tailed distribution of political interactions.

E. Efficiency and Scalability at Archive Scale

A major contribution of this work is the ability to process journalistic archives at scale without relying on costly computational overhead. The full corpus processing encompassed 50,000 news articles, yielding 1,983,499 NER mentions.

By implementing a compiled multi-pattern matcher, the relation extraction throughput increased by approximately 7 $\times$, scaling from 11.6 to 80.8 documents per second. Furthermore, the graph persistence layer was heavily optimized: utilizing a bulk insert strategy into DuckDB accelerated the graph insertion rate by 8 $\times$, improving from 193 to 1,586 edges per second. This architectural efficiency reduces the graph persistence time for the 71,396 extracted edges from over 6 minutes to merely 45 seconds, validating the feasibility of real-time temporal knowledge graph construction.

V. CONCLUSION

This study successfully developed and evaluated a scalable methodological pipeline for constructing a Temporal Knowledge Graph of Peruvian political interactions from a massive

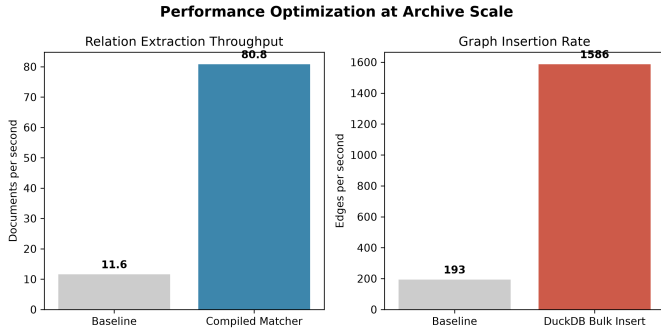


Fig. 4. Throughput improvements achieved by the proposed optimizations. The compiled multi-pattern matcher (left) and DuckDB bulk insertion (right) yielded a 7 $\times$ and 8 $\times$ performance increase, respectively.

journalistic corpus, thereby fulfilling the general objective of the research. The specific objectives were systematically achieved: the massive 50,000-document corpus was successfully preprocessed; a highly optimized entity extraction engine yielding over 1.9 million mentions was implemented; stable relationship categories were organically induced via agglomerative semantic clustering; and the system’s scalability was quantitatively validated.

By achieving these objectives, the study provides clear answers to the established research questions. Regarding RQ1, the experimental results demonstrate that it is highly feasible to build a full-scale graph using open extraction while maintaining exceptional throughput. The implementation of a compiled multi-pattern matcher paired with bulk database insertions allowed for the processing of over 80 documents per second and the writing of 1,586 edges per second, completely overcoming the bottlenecks of traditional classification methods.

Regarding RQ2, the semantic clustering phase proved that it is possible to map political ecosystems without rigidly predefined schemas. The unsupervised approach successfully distilled the raw connecting verbs into 453 coherent semantic clusters. Revealing a heavy-tailed distribution where the top 30 clusters govern nearly 80% of the interactions, the methodology organically discovered nuanced families such as *governance*, *support*, and *denunciation*. Ultimately, the system yielded an interactive, timestamped graph proving to be a highly effective and scalable data intelligence tool for complex temporal analysis.

VI. FUTURE WORK

Although the proposed unsupervised pipeline demonstrates high efficacy in extracting and typing temporal relations, the implementation process revealed several technical and methodological constraints that warrant further investigation.

A. Identified Limitations

- **Single-Source Editorial Bias:** The current corpus is constructed exclusively from the state-owned Agencia Andina. Consequently, the extracted Knowledge Graph inherently inherits any editorial bias, underreporting, or

framing preferences present in government-aligned media, potentially skewing the representation of opposition actors or informal power dynamics.

- **Rigidity in Entity Resolution and Coreference:** The inverse index blocking algorithm implemented for entity resolution is highly efficient but structurally rigid. It struggles with cross-document coreference resolution—for instance, when a politician is referred to exclusively by their title (e.g., “the Minister of Economy”) rather than their named entity, leading to fragmented nodes or lost edges.
- **Evaluation Scalability:** The current evaluation relies on a manually annotated *Gold Standard* of limited size. The lack of a massive, pre-annotated Spanish political dataset makes it difficult to train fully supervised models or dynamically benchmark new iterations of the extraction pipeline without heavy human intervention.

B. Future Work and Open Questions

To address these limitations and expand the analytical capabilities of the system, the following avenues for future research are proposed:

- **Corpus Diversification and Real-Time Streaming:** To construct a balanced and objective Knowledge Graph, the ingestion engine must be expanded to include a diverse spectrum of independent, private, and regional news outlets. Furthermore, transitioning the architecture from a static batch-processing pipeline to a real-time streaming framework (e.g., using Apache Kafka) would allow the graph to update dynamically as news breaks.
- **Deep Entity Linking:** The entity resolution module can be significantly enhanced by replacing the token-based blocking with deep-learning-based Entity Linking (EL). Grounding extracted actors against established structured knowledge bases, such as Wikidata, would resolve complex aliases and implicitly merge synonymous titles into unified nodes.
- **Predictive Analytics via Graph Neural Networks (GNNs):** The current output is a descriptive temporal graph. An open question is whether the generated topology can be used for predictive modeling. Future work could apply Graph Neural Networks (GNNs) or link prediction algorithms to the temporal edges to forecast political alliances, ruptures, or coalition shifts before they are explicitly confirmed in the press.

VII. ETHICAL IMPLICATIONS

The deployment of automated political interaction extraction systems carries significant ethical considerations that must be addressed to prevent the misuse of generated knowledge.

A. Extraction Errors and Defamation

A critical risk involves the potential for extraction errors, where the NLP parser could misclassify neutral co-occurrences as inflammatory relationships due to complex sentence structures. In the high-stakes domain of politics, such automated

assertions can lead to severe reputational damage, the spread of misinformation, or the unfair targeting of public figures. To mitigate this, the system must not be positioned as an autonomous oracle. It is imperative to enforce a "human-in-the-loop" framework, where the generated graph serves strictly as an exploratory data visualization tool that requires rigorous journalistic verification before its insights are published or acted upon.

B. Data Bias and Normalization

The model inherently inherits the sociopolitical biases present in the journalistic source material. If the input news agency systematically underreports specific actors or utilizes biased framing, the resulting Knowledge Graph will statistically reflect and amplify those prejudices as "objective" network patterns. Users must be explicitly warned that the graph is a reflection of a specific corpus's narrative, not an absolute record of political reality.

C. Risk of Automated Surveillance and Profiling

If scaled, the automated extraction of political interactions could facilitate the profiling of activists, political dissidents, or minority groups. By mapping the social and institutional connections of individuals, the system could inadvertently provide a roadmap for the systemic surveillance of vulnerable groups. To prevent this, the deployment of this technology should be restricted to academic and investigative journalism purposes, with strict safeguards against the integration of personal data that could lead to the identification of individuals who are not public figures.

D. Transparency vs. Black-Box Models

Unlike "black-box" approaches that rely on uninterpretable neural networks or proprietary LLMs to classify relationships, our OpenIE approach explicitly extracts the verbatim verb from the text. This prioritizes explainability (XAI). By forcing the system to retain the original predicate and the exact text fragment as evidence for every edge created in the graph, we empower the user to audit the system's logic, transforming the tool into an interpretable analytical aid that encourages critical thinking.

VIII. REFERENCES

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